

Task sheet Week 8: How to read a research paper? k232001

Instructions: Select one of the research papers (Empirical study not a review article) you used for assignment 1. Read the Introduction section of the paper carefully to identify how the author(s) develop the research problem by locating the following elements:

- ⇒ **Context** – Background information about the topic or research area
- ⇒ **Importance** – Why the issue is significant or needs attention
- ⇒ **Research Gap** – What limitation, missing area, or unresolved issue exists in previous research
- ⇒ **Purpose of the Study** – What the authors aim to investigate or achieve

Task 1: Complete the table below using sentences or summarized ideas from the paper.

	Explanation in your words (Use paraphrase technique)	Evidence(sentence from the paper)
Context	The rapid advancement and widespread public availability of Large Language Models (LLMs) enable users to easily generate high-quality text and programming code, leading students to increasingly depend on these tools to complete their educational coursework.	"In recent years, an increase in capability has been observed in the development of LLMs, enabling them to generate high-quality, co-herent paragraphs, answer questions, and even produce human-like code."
Importance	The growing reliance on AI tools has caused a surge in academic dishonesty, forcing educators to use AI-generated content detectors to check student work; if these detectors are flawed, it actively threatens the integrity of educational assessments and can result in unjust grading for students.	"This discrepancy can lead to disparities in the evaluation of students' academic submissions, potentially resulting in unfair grading."
Research Gap	Even though current AI detectors have shown success in catching AI-written essays and natural language text, it is still unknown whether they can reliably detect AI-generated programming code, because code has a much more rigid and complex structure than regular text.	"While existing AIGC Detectors have proven their proficiency in identifying AI-generated text, their effectiveness in recognizing AI-generated code remains uncertain due to the intricate nature of programming code."
Purpose of study	The authors aim to perform a data-driven experiment to test how accurately various existing AI detectors can identify AI-generated code, especially when the code features different structural modifications and prompt variations.	"The paper aims to conduct an empirical study to evaluate the performance of different AIGC Detectors in detecting AIGC across diverse contextual and syntactical variations."

Task 2: Based on your analysis above, write a clear research problem statement that summarizes the issue the selected study addresses:

Research Problem Statement (2-3 sentences)

As students increasingly utilize accessible Large Language Models to generate programming solutions, educators are forced to rely on AI content detectors to prevent academic dishonesty. However, while these detectors are proven to work on natural language, their effectiveness at identifying the complex syntax of AI-generated code remains unverified, creating a significant risk of unfair grading. Therefore, this study addresses the urgent need to empirically evaluate the performance and limitations of current AI detectors when applied to various contextual and syntactical code variations